

Day 12: Essential DAX Aggregation & Logical Functions

[DEF] Definition to Remember

Essential DAX functions provide standard mathematical aggregations (SUM, AVERAGE, COUNT), safe division (DIVIDE), and conditional branching logic (IF, SWITCH).

1. Core Mathematical & Aggregation Functions

```
// Sum and Average
Total Revenue = SUM(Fact_Sales[Revenue])
Average Order Value = AVERAGE(Fact_Sales[Revenue])

// Row Counts
Total Orders = COUNT(Fact_Sales[OrderID])
Unique Customers = DISTINCTCOUNT(Fact_Sales[CustomerID])
Total Rows = COUNTROWS(Fact_Sales)
```

2. Safe Mathematical Division: `DIVIDE()`

Always use DIVIDE() instead of the standard / operator to eliminate divide-by-zero crashes:

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)

3. Conditional Logical Branching: `IF()` vs `SWITCH()`

```
// Using IF for single condition
Discount Tier =
IF(
    [Total Sales] > 100000,
    "Platinum",
    "Standard"
)

// Using SWITCH(TRUE(), ...) for multi-condition evaluation (Cleaner than nested IFs)
Customer Performance Rating =
SWITCH(
    TRUE(),
    [Total Sales] >= 500000, "Tier 1: Enterprise Leader",
    [Total Sales] >= 100000, "Tier 2: Key Strategic Account",
    [Total Sales] >= 25000, "Tier 3: Growth Customer",
    "Tier 4: Standard Customer"
)
```

[MUST-WRITE] Must-Master Skills & Interview Points

- 1. Why DIVIDE(A, B, 0) is safer and faster than IF(B = 0, 0, A / B).
- 2. Difference between COUNT() (ignores blanks), COUNTROWS() (table row count), and DISTINCTCOUNT().
- 3. Mastering SWITCH(TRUE(), ...) for clean multi-tier business logic.

[RECALL] Quick Recall

SUM / AVERAGE / DISTINCTCOUNT + DIVIDE(Safe Division) + SWITCH(TRUE(), conditions)